Solar ESS Co. Ltd.
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Tel 063 151 6694, 089 632 5829
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DAILY CONSUMPTION · SolisCloud · 20 – 22 October 2026

Ban Doo Chiangrai: when the site uses its electricity

20 kWp PV · Solis 20 kW three-phase hybrid inverter, no battery · zero export · small factory with a 57–62 kW machine · sunrise 06:17, sunset 17:53

≈ 20 kWh
used after sunset on 20 Oct, all bought from PEA

≈ 80 kWh
used after 17:30 on 21 Oct (machine day)

8.8 kW
being exported at 09:05 on 22 Oct with no load on site

07:30–16:30
suggested working hours to follow the sun

Three days side by side

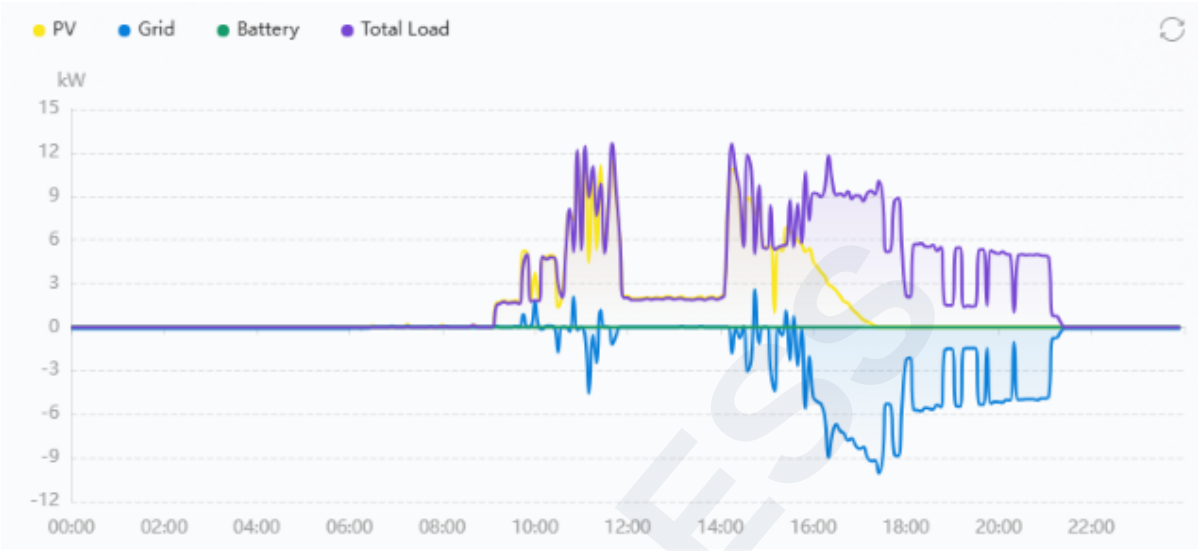
Day	Type	Solar produced	Site load	Bought from PEA	Exported	When the load runs
Tue 20 Oct	Normal day	≈ 32 kWh	≈ 68 kWh	≈ 36 kWh	1.5 kWh	09:30 – about 21:00; 5–7 kW after sunset
Wed 21 Oct	Machine day	47.3 kWh	281 kWh	230 kWh	5.2 kWh	Machine bursts of 50–60 kW from 09:00 to about 20:30
Thu 22 Oct (to 09:05)	Morning	6 kWh so far	0 kW shown	–	8.8 kW now	No load yet; the surplus is going to the grid

Solar produced = daily full-load hours × 20 kWp, which also matches the daily earning at 4.7 THB/kWh. 20 October load and purchases are read from the SolisCloud day view and are approximate. 21 October figures come from the inverter's 5-minute data.

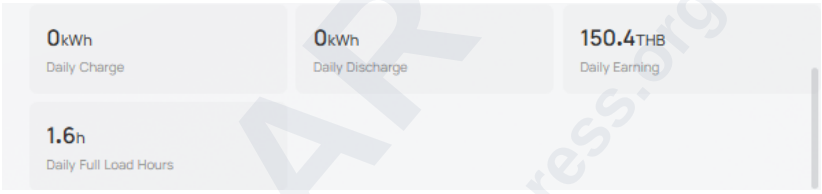
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Tuesday 20 October · a normal day



SolisCloud day view, 20 October. Purple = site load, yellow = solar, blue = grid (below zero = bought from PEA).

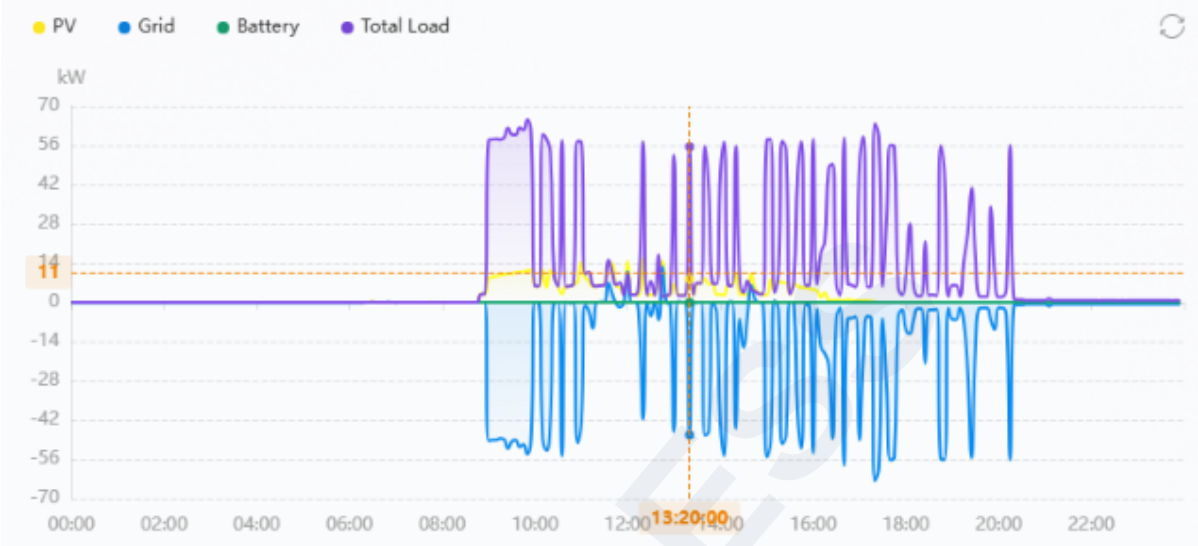


Daily totals: 1.6 full-load hours (≈ 32 kWh), 150.4 THB earned.

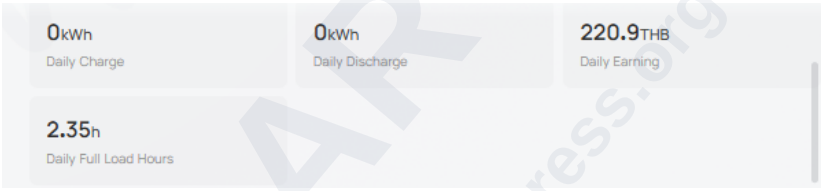
- **Morning wasted.** The sun is up from 06:17, but the load only starts around 09:30, so about two hours of solar are not used.
- **Lunch dip.** Between about 12:30 and 14:30 the load drops to 2–3 kW and the inverter is held back with it.
- **Evening on the grid.** From 17:30 until about 21:00 the site keeps drawing 5–7 kW after sunset. That is roughly 20 kWh, all bought from PEA – more than half of the day's purchases.

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Wednesday 21 October · a machine day



SolisCloud day view, 21 October. The 57–62 kW machine runs in bursts all day and into the evening.



Daily totals: 2.35 full-load hours (≈ 47 kWh), 220.9 THB earned.

- **Solar at full output while the machine runs** – about 10–13 kW – but the machine draws four to five times that, so most of each burst comes from PEA.
- **Late running.** The machine keeps cycling until about 20:30. About 80 of the day's 281 kWh (29%) were used after 17:30, when there is no sun at all.
- **Limit slow to release.** On this day the export limit took 45–55 minutes to release after the machine started, so solar was held back during part of the run.

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Thursday 22 October · 09:05, morning



- **Producing with no load.** The panels are making 8.9 kW, the load shows 0 kW, and 8.8 kW is going to the grid ("Exporting").
- **Different from the rest of October,** when the export limit kept the inverter idle until the shop opened. Either the export limit has been switched off, or the meter/CT reading has dropped out (a load of exactly 0 kW is unusual for this site).
- **Exported energy is not paid for.** Please check Allow Export = Disable and the meter connection.

What these days show

> **The site uses most of its electricity when the sun is low or gone.** The panels produce best between about 08:00 and 16:00. The site starts around 09:30, slows at lunch, and keeps working until about 21:00 – so the morning sun is wasted and the evening is bought from PEA.

Suggestion	Possible gain / month	Possible saving / month
Move office and working hours to follow the sun, for example 07:30–16:30 instead of about 09:30–21:00	up to ~500 kWh	up to ≈ 2,400 THB
Run the machine between 09:00 and 15:00 where the work allows, and avoid runs after 17:30		adds to the gain above
Stagger lunch breaks so some load keeps running between 12:30 and 14:30		uses solar that is now curtailed
Check the export setting and meter link after this morning's export		stops unpaid export

Estimates at 4.7 THB/kWh. The gain from shifting hours assumes the ~20 kWh used after sunset on a normal day moves into sun hours on about 26 working days a month; it overlaps with the other daytime-load suggestions in the production timeline report. Sources: SolisCloud plant overview and day views, 20–22 October 2026, and inverter 5-minute data for 21 October. Analysis by Solar ESS Co. Ltd.